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Department of Economics

**M.Sc. in Business Administration, Analytics
and Information Systems**

**Probabilistic Machine Learning Models and
Decision Optimization for Hepatocellular Carcinoma Diagnosis**

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The Problem & The Goal

- **Hepatocellular Carcinoma (HCC):**
A major cause of cancer-related mortality worldwide.
Early diagnosis is a decisive for prognosis.
- **The Gap:**
There is a disconnect between accurate risk prediction and clinical decision-making.
Many studies stop at model selection and apply arbitrary thresholds (e.g., 0.50).
- **Thesis Goal:**
To develop a framework that transforms calibrated HCC risk probabilities into clinically and economically justified decision thresholds.



Part I: Data & Probabilistic ML

- **Source:** Synthetic, but medical realistic dataset (Kaggle).
- **Size:** 5,000 observations, 14 variables.
- **Target:** HCC Presence (21.8%) vs Absence (78.2%).
- **Feature categories:**



Demographics

- Age
- Gender
- BMI



Lifestyle

- Alcohol
- Smoking
- Physical Activity



Medical History

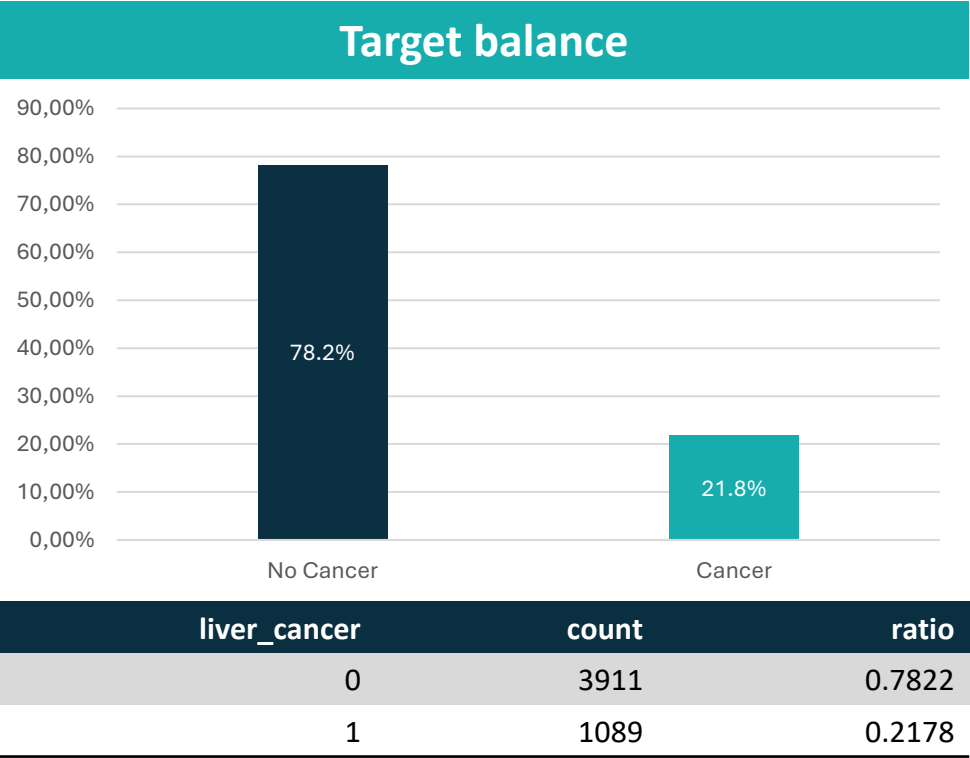
- Hepatitis B & C
- Cirrhosis
- Diabetes
- Family History



Biomarkers

- AFP Level
- Liver Function

Dataset Overview	
Metric	Value
Total Samples	5,000
Number of Features	14
Target Prevalence	21.8%

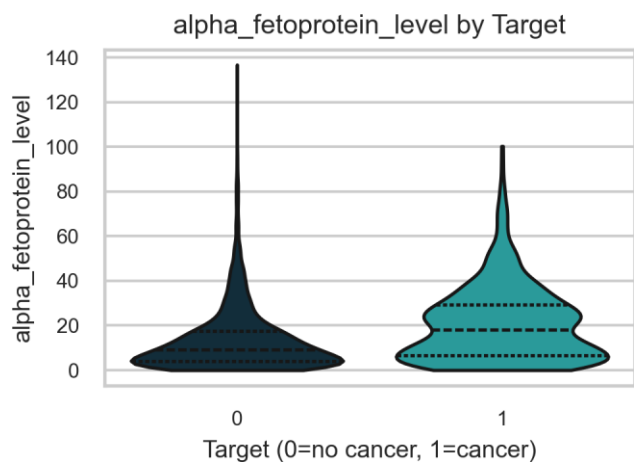
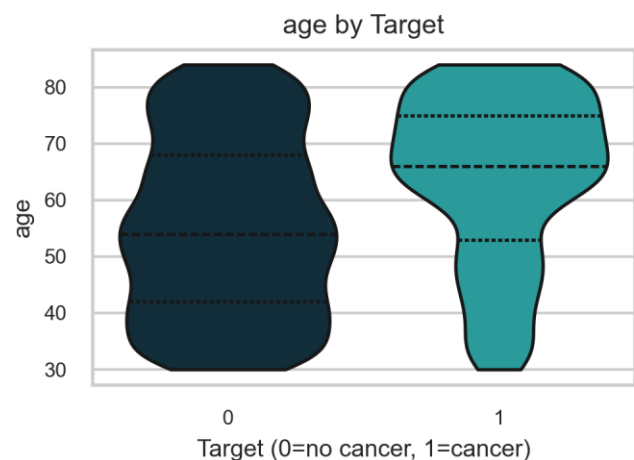


Dataset Dimensions and Data Types	
Shape (rows, cols): (5000, 14)	
Dtypes:	
age	int64
gender	object
bmi	float64
alcohol_consumption	object
smoking_status	object
hepatitis_b	int64
hepatitis_c	int64
liver_function_score	float64
alpha_fetoprotein_level	float64
cirrhosis_history	int64
family_history_cancer	int64
physical_activity_level	object
diabetes	int64
liver_cancer	int64
dtype: object	

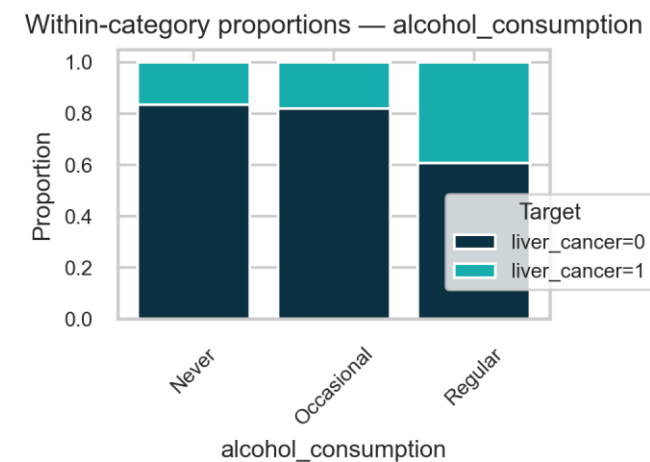
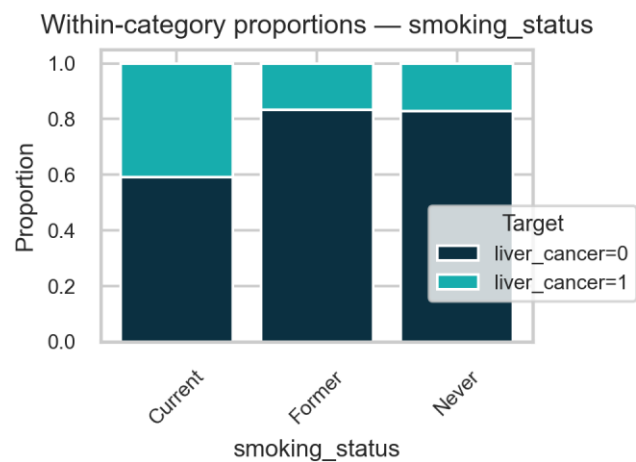
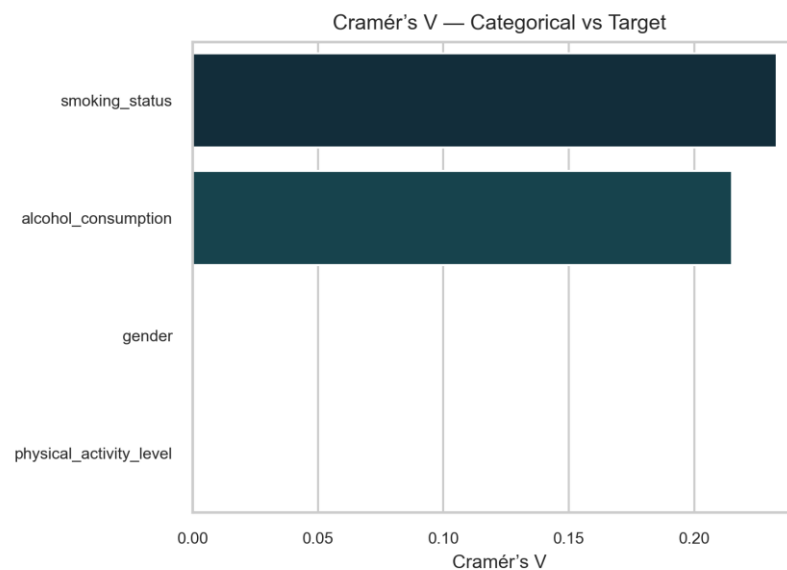
Exploratory Data Analysis (EDA)

- **Key Findings:**
 - **Age:** Clear shift towards older ages for positive cases.
 - **AFP:** Strong separation, significantly higher values in cancer cases.
 - **Risk Factors:** Smoking and alcohol consumption show the strongest associations among categorical variables (Cramer's $V > 0.2$).
- **Correlations:** Cirrhosis and diabetes exhibit the highest point-biserial correlations with the target.

Age & AFP by Target



Cramer's V Association Strength & Within-Category Proportions



Machine Learning Methodology

- **Data Split:**
 - Train (60%) / Validation (20%) / Test (20%).
 - Stratified split to maintain class proportions.
- **Preprocessing:** Imputation, One-Hot Encoding and Scaling within a leakage-free pipeline.
- **Optimization:** GridSearchCV (5-fold CV) maximizing ROC-AUC.

Logistic Regression
(LR)

Baseline,
Linear Model

Random Forest
(RF)

Ensemble, non-Linear,
Reduces Variance

XGBoost
(XGB)

Gradient Boosting,
High Performance

Model Results (Test Set)

- **Performance Comparison:**
 - XGB achieved near-perfect discrimination ($AUC \approx 1.0$, $AP \approx 1.0$) on the synthetic dataset.
 - RF followed with $AUC \approx 0.99$.
- **Part I Conclusion:** XGB was selected as the optimal model for the decision optimization in Part II.

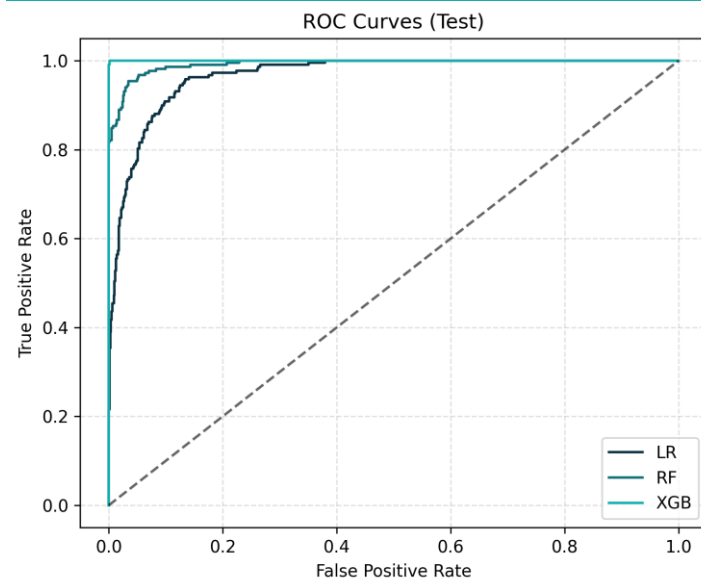
Test Set Summary Table					
model	auc	auc_ci	ap	ap_ci	brier
XGB	1.0000	(0.9999, 1.0)	1.0000	(0.9998, 1.00)	0.0013
RF	0.9926	(0.9884, 0.9962)	0.9788	(0.9677, 0.9883)	0.0273
LR	0.9671	(0.9571, 0.9763)	0.8961	(0.8619, 0.9267)	0.0593

Robustness Diagnostics (Part 1)

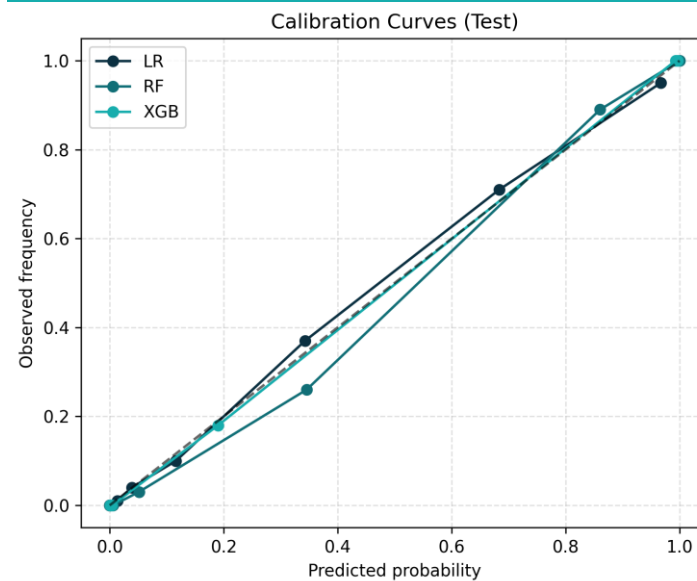
- **Sanity Checks:** Performed to rule out artifacts due to high performance.
 - Label Permutation: Performance dropped to chance level (0.5), confirming no data leakage.
- **Biomarker Ablation (Drop AFP & LFS):**
 - Removing strong biomarkers drops AUC to 0.936.
 - Conclusions: The model exploits strong biomarker signals but retains high predictive power based on risk factors alone.

Feature Ablation Stress Test Summary Table						
Model	CV Best ROC-AUC	Validation Brier	Test ROC-AUC	Test AP	Test Brier	
LR	0.9259	0.0799	0.9309	0.8404	0.0780	
RF	0.9282	0.0708	0.9316	0.8454	0.0762	
XGB	0.9350	0.0660	0.9362	0.8580	0.0685	

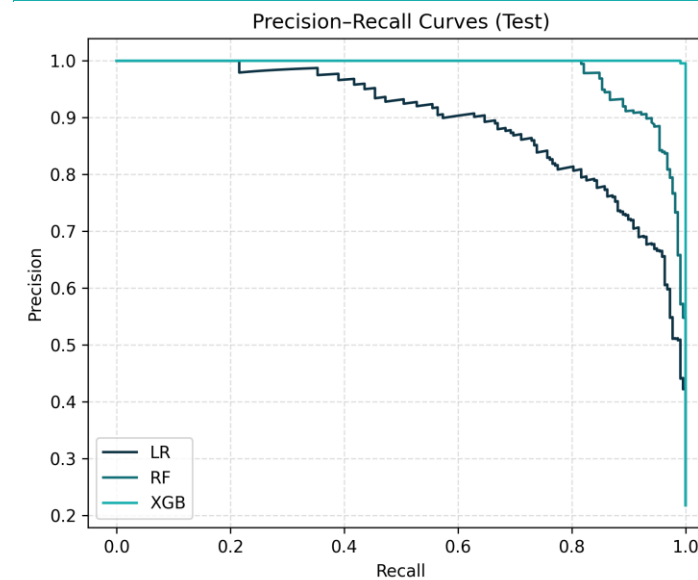
ROC Curves



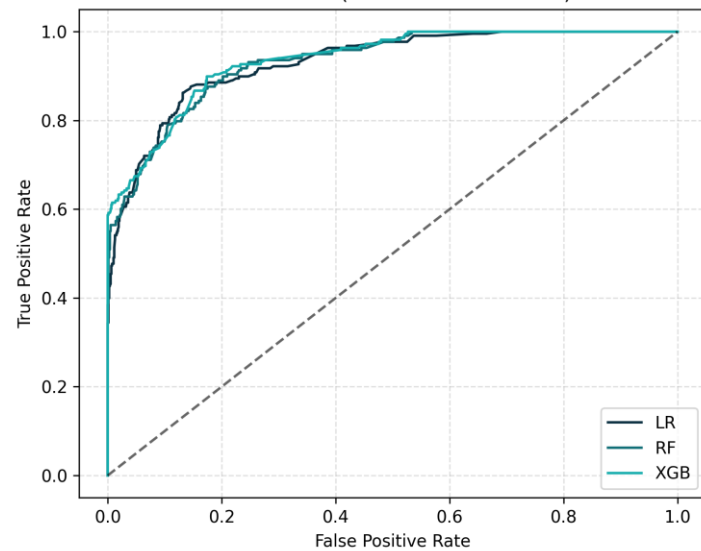
Calibration Curves



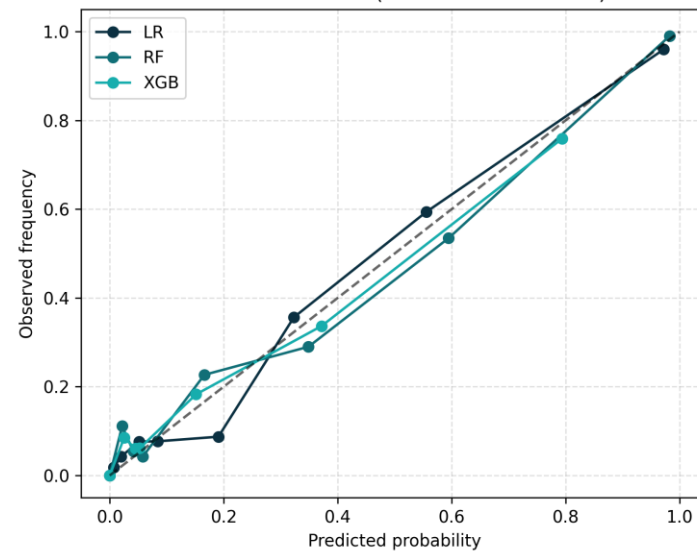
PR Curves



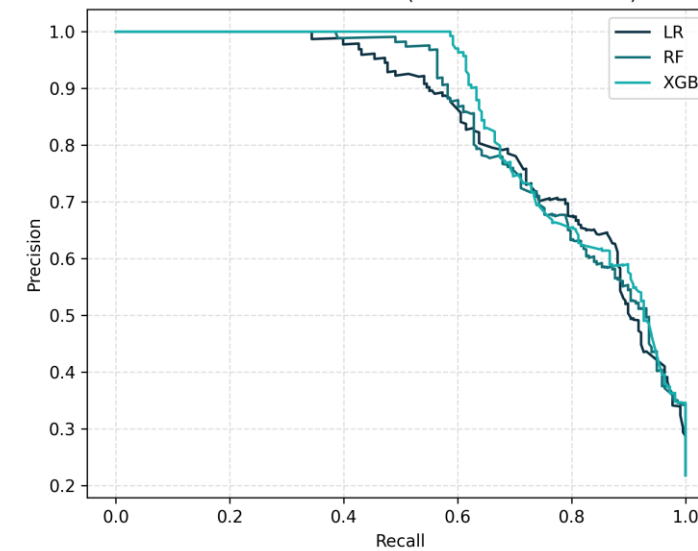
ROC Curves (Stress Test AFP & LFS)



Calibration Curves (Stress Test AFP & LFS)



Precision-Recall Curves (Stress Test AFP & LFS)

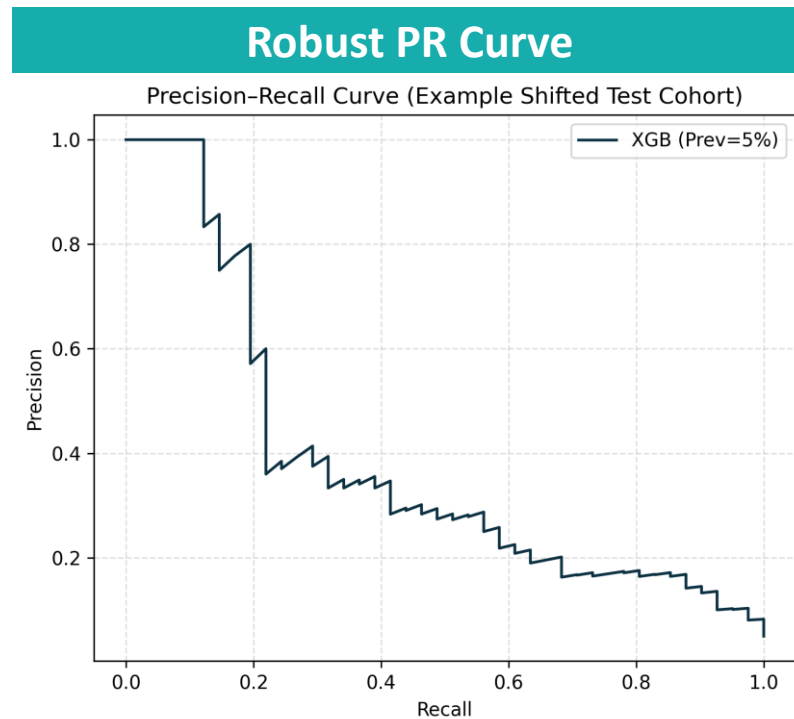


Robustness Diagnostics (Part 2)

Stress Tests:

- **Noise Injection:** Adding noise reduced AUC to 0.86, indicating that perfect performance is driven by the cleanness of synthetic data.
- **Prevalence Shift:** In low-prevalence setting (5%), Precision is impacted more significantly than AUC.

Robust Test Set Results			
model	AUC	AP	Brier
LR	0.8626 [0.8234, 0.8997]	0.3732 [0.2245, 0.4850]	0.0640 [0.0577, 0.0713]
RF	0.8733 [0.8367, 0.9043]	0.3845 [0.2411, 0.4966]	0.0617 [0.0575, 0.0676]
XGB	0.8825 [0.8476, 0.9098]	0.4071 [0.2596, 0.4874]	0.0606 [0.0561, 0.0678]



Part II: Decision Optimization

- **The Question:** “Given a predicted probability P , when should the clinician escalate?”
- **Decision Theory:** Pauker – Kassirer threshold model.
 - Escalate if $P \geq p^*$.
 - p^* depends on the utility/cost of outcomes (TP, FP, FN, TN).
- **Calculation Formula:**

$$p^* = \frac{L_{FP}}{G_{TP} + L_{FP}}$$

Where L_{FP} the Net Loss of FP and G_{TP} the Net Benefit of TP.

Clinical Scenarios (QALYs)

- **Unit:** Quality Adjusted Life Years (QALYs).
- **Scenario 1(Balanced):** Based on literature values.
 $p^* = 35.7\%$
- **Scenario 2 (Conservative – Risk Averse):** High weight on avoiding FN (missing cancer).
 $p^* = 6.2\%$
- **Scenario 3 (Aggressive):** High weight on avoiding FP (unnecessary work-up).
 $p^* = 73.5\%$

Payoff Matrix – Clinical Scenarios							
Scenario	Approach	Risk Attitude	U_{11} (TP)	U_{10} (FN)	U_{01} (FP)	U_{00} (TN)	p^*
1	Clinical (QALYs)	Balanced	-0.060	-0.130	-0.039	0.000	35.7%
2	Clinical (QALYs)	Conservative	-0.060	-0.650	-0.039	0.000	6.2%
3	Clinical (QALYs)	Aggressive	-0.060	-0.130	-0.195	0.000	73.5%

Economic Scenarios (Cost-Benefit Analysis)

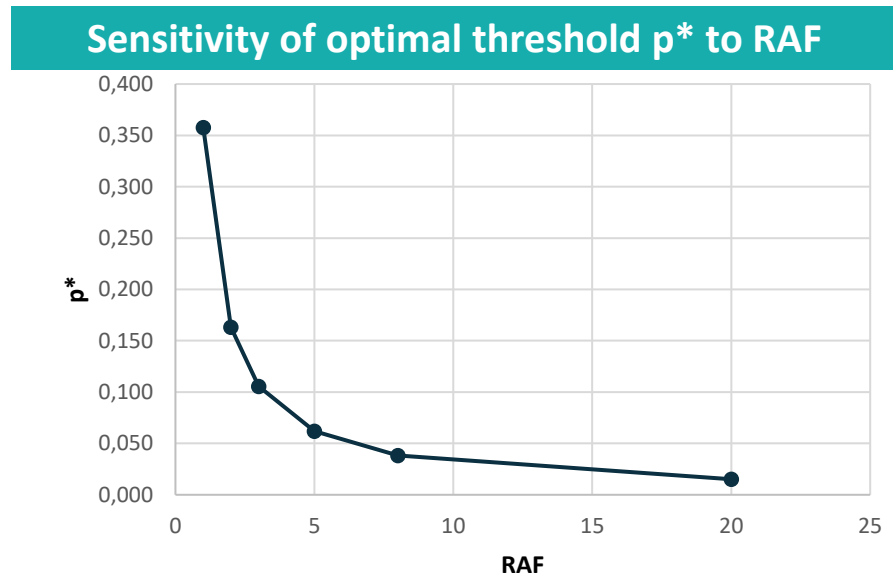
- **Unit:** Cost in USD.
- **Scenario 4 (Balanced):** Direct medical costs.
 $p^* = 3.8\%$
- **Scenario 5 (Societal – Conservative):** Includes indirect costs, very low threshold.
 $p^* = 1.4\%$
- **Scenario 6 (Cost-Containment):** Penalizes overuse.
 $p^* = 44.4\%$

Payoff Matrix – Economic Scenarios							
Scenario	Approach	Risk Attitude	U_{11} (TP)	U_{10} (FN)	U_{01} (FP)	U_{00} (TN)	p^*
4	Economic (USD)	Balanced	-62775	-92133	-1171	0	3.8%
5	Economic (USD)	Conservative	-177658	-260737	-1171	0	1.4%
6	Economic (USD)	Aggressive	-62775	-92133	-23420	0	44.4%

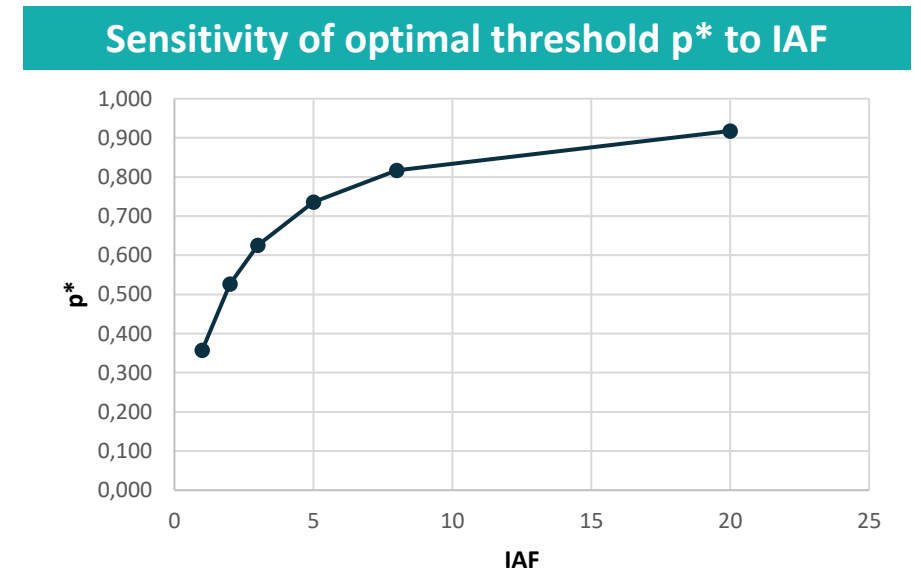
Sensitivity Analysis

- **Purpose:** How does p^* change when decision-maker preferences shift.
- **Parameters:**
 - RAF (Risk Aversion Factor): Aversion to FN.
 - IAF (Intervention Averse Factor): Aversion to FP.
 - OM (Overuse Multiplier): Penalizes overuse.
- **Conclusions:** The optimal threshold varies continuously and non-linearly. It is a policy choice, not a model constant.

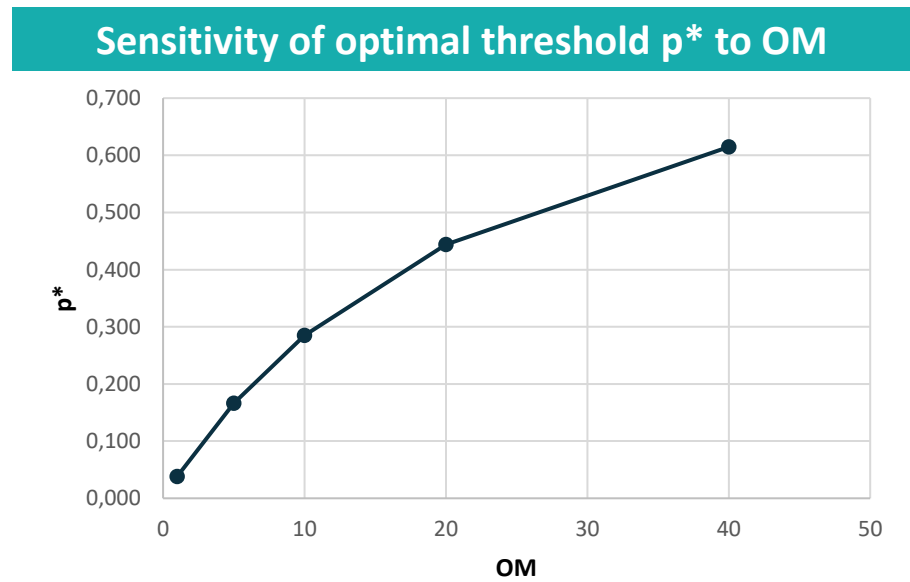
Sensitivity Table RAF					
Multiplier	Value	U_{10}	p^*	$p^* (\%)$	
RAF	1	-0.130	0.358	35.78%	
RAF	2	-0.260	0.163	16.32%	
RAF	3	-0.390	0.106	10.57%	
RAF	5	-0.650	0.062	6.20%	
RAF	8	-1.040	0.038	3.83%	
RAF	20	-2.600	0.015	1.51%	



Sensitivity Table IAF					
Multiplier	Value	U_{01}	p^*	$p^* (\%)$	
IAF	1	-0.039	0.358	35.78%	
IAF	2	-0.078	0.527	52.70%	
IAF	3	-0.117	0.626	62.57%	
IAF	5	-0.195	0.736	73.58%	
IAF	8	-0.312	0.817	81.68%	
IAF	20	-0.780	0.918	91.76%	



Sensitivity Table OM					
Multiplier	Value	U_{01}	p^*	$p^* (\%)$	
OM	1	-1171	0.038	3.84%	
OM	5	-5855	0.166	16.63%	
OM	10	-11710	0.285	28.51%	
OM	20	-23420	0.444	44.37%	
OM	40	-46840	0.615	61.47%	



Operationalization

- Applying thresholds to the test set.
- **Observations:** Although thresholds vary widely (1.4% to 73.5%), escalation rates cluster near the prevalence (21.8% - 23.8%).
- **Reason:** The model is highly confident (well-calibrated and discriminatory), so most probabilities are near 0 or 1. Few cases fall into the grey zone.

Escalation Rate & Payoff Matrix								
Scenario	p*	Escalate_n	Escalate_%	TP	FP	FN	TN	
S5_Econ_Conservative	0.0139	238	0.238	218	20	0	762	
S4_Econ_Balanced	0.0384	226	0.226	218	8	0	774	
S2_Clinical_Conservative	0.0620	224	0.224	218	6	0	776	
S1_Clinical_Balanced	0.3578	220	0.220	218	2	0	780	
S6_Econ_Aggressive	0.4437	220	0.220	218	2	0	780	
S3_Clinical_Aggressive	0.7358	218	0.218	217	1	1	781	

Operational Metrics							
Scenario	Sensitivity	Specificity	PPV	NPV	EU_per_patient	EU_total	
S5_Econ_Conservative	1.0000	0.9744	0.92	1	-38752.8640	-3.8753E+07	
S4_Econ_Balanced	1.0000	0.9898	0.96	1	-13694.3180	-1.3694E+07	
S2_Clinical_Conservative	1.0000	0.9923	0.97	1	-0.0133	-1.3314E+01	
S1_Clinical_Balanced	1.0000	0.9974	0.99	1	-0.0132	-1.3158E+01	
S6_Econ_Aggressive	1.0000	0.9974	0.99	1	-13731.7900	-1.3732E+07	
S3_Clinical_Aggressive	0.9954	0.9987	0.995	0.9987	-0.0133	-1.3345E+01	

Limitations

- **Synthetic Data:** Near-perfect performance is likely due to the lack of noise/complexity found in real clinical data.
- **Generalizability:** External validation on real-world data is required.
- **Economic Data:** Based on U.S. estimates, may differ in other healthcare systems.
- **Binary Decision:** Clinical practice is often sequential, not a single one-off decision.

Conclusions & Future Work

- **Conclusions:**
 - High accuracy (AUC) is not enough. Calibration and consequence-aware thresholds are required.
 - The optimal threshold is a policy parameter.
- **Future Work:**
 - Application to real-world data.
 - Probabilistic Sensitivity Analysis (PSA) for uncertainty.
 - Modeling multi-stage decision pathways.

Thank You!

Questions & Answers

Happy to discuss methods, calibration and decision thresholds.

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Derivation of p^*

Payoff Matrix		
Reality / Decision	Escalate ($\delta = 1$)	No Escalation ($\delta = 0$)
Cancer ($Y = 1$)	U_{11} (TP)	U_{10} (FN)
No Cancer ($Y = 0$)	U_{01} (FP)	U_{00} (TN)

- $\mathbb{E}[U_{\text{escalate}}|P] = P \cdot U_{11} + (1 - P) \cdot U_{01}$
- $\mathbb{E}[U_{\text{no escalation}}|P] = P \cdot U_{10} + (1 - P) \cdot U_{00}$
- $p^* = \frac{U_{00} - U_{01}}{(U_{11} - U_{10}) + (U_{00} - U_{01})}$
- $L_{\text{FP}} = (U_{00} - U_{01})$ & $G_{\text{TP}} = (U_{11} - U_{10})$

Calibration Selection

Calibration Selection Table		
model	picked	val_brier
LogisticRegression	sigmoid	0.0587
RandomForest	isotonic	0.0276
XGBoost	sigmoid	0.0010

- Because Part II uses probabilities rather than simple classifications, it is not enough for the model to have a good AUC.
- The probabilities need to be reliable. This is why post-hoc calibration and selection based on validation Brier score were performed.

Leakage & Sanity Checks

Label Permutation

XGB baseline CV AUC (mean±std): 0.9992206018402776 0.0008799416177036603

XGB label-permutation CV AUC (mean±std): 0.5011236080492399 0.017120018599074048

Repeated Splits

XGB repeated DEV splits AUC (mean±std): 0.9998576989460541 0.00041389135344267546

XGB repeated DEV splits AUC (min/max): 0.9981179538026514 1.0

- The thesis is not saying that the model is clinically perfect. It is saying that within this synthetic dataset there is no evidence of obvious leakage.
- The near-perfect performance is mainly explained by the clean synthetic structure of the data.

Feature Ablation Details

Feature Ablation Stress Test XGB				
Scenario	XGB Test ROC-AUC	XGB Test AP	XGB Brier	
Full Model	~1.0000	~1.0000	0.0013	
Drop AFP	0.9669	0.9110	0.0506	
Drop LFS	0.9743	0.9380	0.0374	
Drop AFP & LFS	0.9362	0.8580	0.0685	

- AFP and Liver Function Score are strong predictors, but they are not the only ones.
- When both are removed, the performance drops, but remains well above chance level. So, the model also considers other risk factors.

Utility Assumptions

Clinical Utility

- **Unit:** QALYs
- **TP:** early detection of HCC
- **FN:** delayed diagnosis and progression to advanced stage
- **FP:** unnecessary work-up escalation
- **TN:** reference outcome

Utility Assumptions

Economic Utility

- **Unit:** USD
- **TP:** cost of escalation with benefit of early detection
- **FN:** cost at an advanced stage because of a missed escalation
- **FP:** loss due to unnecessary work-up
- **TN:** reference outcome